Report 2: Measuring Internal Resistance (Rm) of Moving Iron Instrument (DC)

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1. Objective

The aim of this experiment is to calculate the internal resistance (Rm) of a moving iron voltmeter by analyzing the change in voltage across it when known resistances (Rs) are added in series with a DC voltage supply.

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2. Theory

When a moving iron voltmeter with internal resistance RmR_mRm is placed in series with an external resistor RsR_sRs and connected to a DC power source of voltage EEE, the voltage VVV across the instrument is governed by the following relationship:

$$V = E \cdot \frac{R_m}{R_m + R_s}$$

$$\frac{E}{V} = \frac{R_m + R_s}{R_m} \Rightarrow \frac{E}{V} = 1 + \frac{R_s}{R_m}$$

$$Y = \frac{E}{V} - 1$$

$$Y = \frac{1}{R_m} \cdot R_s$$

This equation is in the standard form of a straight-line equation: Y=mX+cY = mX + cY=mX+c, where:

- YYY is the dependent variable
- RsR_sRs is the independent variable
- m=1Rmm = \frac{1}{R_m}m=Rm1 is the slope
- ccc is the intercept (ideally zero)

By plotting YYY against RsR_sRs, the value of RmR_mRm can be obtained as the reciprocal of the slope of the best-fit line.

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4. Experimental Data

Series Resistance (R_s) (Ohms)	Measured Voltage (V_{dc}) (Volts)
0	30.0
10	28.8
20	27.0
30	25.6
40	24.8
50	23.5

5. Calculations

Assuming the DC source voltage E_dc = 30V. We calculate Y = (E_dc / V_dc - 1) for each set of readings:

Rs (Ohms)	V_dc (Volts)	E_dc/V_dc	Y = (E_dc/V_dc - 1)
0	30.0	1.000000	0.000000
10	28.8	1.041667	0.041667
20	27.0	1.111111	0.111111
30	25.6	1.171875	0.171875
40	24.8	1.209677	0.209677
50	23.5	1.276596	0.276596

Linear Regression Results:

- **Slope (m = 1/Rm)**: 0.005565
- **Intercept (c)**: -0.003972
- **R^2 Value**: 0.994587 (indicating excellent linear correlation)

Calculated Internal Resistance:

$$R_m = \frac{1}{\text{slope}} = \frac{1}{0.005565} = 179.69\,\Omega$$

The minor non-zero intercept is likely due to slight inaccuracies in measurements or rounding effects.

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6. Graphical Method

To determine RmR_mRm graphically:

1. Plot the calculated values of Y=(EdcVdc−1)Y = \left(\frac{E_{dc}}{V_{dc}} - 1\right)Y=(VdcEdc−1) on the Y-axis.
2. Plot the corresponding series resistance values RsR_sRs on the X-axis.
3. Draw a best-fit straight line through the points.
4. Find the slope of this line.
5. The reciprocal of the slope gives the internal resistance RmR_mRm.

Data for Plotting:

R_s (Ohms)	$Y = (E_{dc}/V_{dc} - 1)$
0	0.000000
10	0.041667
20	0.111111
30	0.171875
40	0.209677
50	0.276596

Expected Slope from Graph: Approximately **0.005565**

7. Conclusion

Through both mathematical analysis and graphical interpretation, the internal resistance RmR_mRm of the moving iron voltmeter was calculated to be **179.69 Ohms**. The high correlation coefficient ($R^2 \approx 0.9946$) confirms the reliability of the method and the consistency of the data.

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8. Note

The determined value Rm=179.69 ΩR_m = 179.69 \, \OmegaRm=179.69Ω will be used as a reference in future experiments involving this moving iron voltmeter.